SOFTWARE REQUIREMENTS SPECIFICATION

Customs Performance Measurement Mechanism (PMM) Dashboard

B2G Customs Intelligence and Performance Analytics Platform

Document Version	2.0
Status	Enhanced Baseline / Implementation Ready
Prepared Date	05 October 2026
Primary Stack	React + TypeScript + Vite + AG Grid / Node.js + Express + TypeScript / MySQL
Data Volume Target	50,000 simulated shipments over 24 months; scalable to 10x prototype volume

Important positioning note

This project is a simulated, WCO-aligned performance analytics application for demonstration, learning, portfolio, and prototype purposes. It does not constitute an official WCO PMM implementation, certification, tariff determination service, or legal customs valuation system.

1. Document Control

Item	Value
Document Title	Software Requirements Specification - Customs Performance Measurement Mechanism (PMM) Dashboard
Document ID	PMM-SRS-001
Version	2.0
Status	Enhanced Baseline / Implementation Ready
Prepared Date	05 October 2026
Primary Audience	Project Owner, Developers, QA Engineers, UI/UX Designers, Technical Reviewers, Customs Domain Stakeholders
System Type	Business-to-Government (B2G) enterprise analytics dashboard
Deployment Model	Web application; cloud-ready

1.1 Revision History

Version	Date	Author/Owner	Change Summary
1.0	05-Oct-2026	Project Team	Initial complete SRS baseline derived from the project concept and current WCO reference material.
2.0	05-Oct-2026	Project Team	Added React + TypeScript + Vite, AG Grid data-grid requirements, PMM-draft enhancements, richer TRS/revenue analytics, organizational metrics, scheduled rules, alert auto-resolution, deterministic simulation, and expanded acceptance criteria.

1.2 Approval / Sign-Off

Role	Name	Status
Project Owner	TBD	Pending
Technical Lead	TBD	Pending
Domain Reviewer	TBD	Pending
QA Lead	TBD	Pending

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3. Introduction and Purpose

3.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the business, functional, data, interface, security, performance, and technical requirements for the Customs Performance Measurement Mechanism (PMM) Dashboard. The document is intended to provide a single implementation baseline for the design and development team and to make the expected behavior of the platform measurable and testable.

This revision additionally standardizes the implementation baseline on React + TypeScript + Vite, Node.js + Express + TypeScript, MySQL, AG Grid, typed API contracts, and the enhanced PMM-draft capabilities described in Section 8 and Section 21.

The solution is designed as a centralized customs intelligence hub that transforms high-volume transactional data into performance indicators, alerts, comparative views, and analytical findings. The platform will emulate the structure and decision-support intent of the World Customs Organization (WCO) Performance Measurement Mechanism while using simulated data and configurable business rules.

WCO alignment

The WCO states that its PMM is structured around four dimensions of Customs performance: Trade Facilitation and Economic Competitiveness; Revenue Collection; Enforcement, Security and Protection of Society; and Organizational Development. The WCO PMM was endorsed by the WCO Council in June 2023. [1]

3.2 Project Context

Customs organizations process large volumes of declarations, cargo movements, inspections, valuations, tariff classifications, and release events. Decision-makers require timely evidence to identify delays, revenue leakage, enforcement anomalies, and organizational weaknesses. This project packages those use cases into an

enterprise-style dashboard so that a user can move from a national KPI view to a port, commodity, shipment, or anomaly level without manually processing raw transactional data.

3.3 Intended Use

- Demonstrate an end-to-end B2G analytics product using realistic customs workflows and relational data.
- Provide executives with national and port-level KPI visibility.
- Provide operational managers with Time-Release Study (TRS) bottleneck analysis.
- Provide revenue users with analytical valuation variance and theoretical-revenue gap analysis.
- Provide enforcement users with inspection, seizure, and anomaly indicators.
- Provide administrators with configurable KPI thresholds, roles, access scopes, and audit visibility.
- Serve as a portfolio-quality demonstration of SQL aggregation, REST APIs, RBAC, dashboard UX, data engineering, and enterprise analytics.

4. Product Overview and Objectives

4.1 Product Vision

The PMM Dashboard will convert customs operational data into a decision-oriented performance layer. The primary user journey is: select a scope and time period -> review KPI status -> identify an outlier -> drill into the responsible port/commodity/shipment -> understand the root cause -> record or communicate the required action.

4.2 Business Objectives

- Reduce the time required to understand national and terminal-level customs performance.
- Make clearance delays measurable and comparable across ports and periods.
- Highlight possible valuation anomalies using a configurable historical reference model.
- Surface enforcement outcomes such as flagged and seized declarations without requiring raw-data inspection.
- Provide role-sensitive access to national, regional, and terminal data.
- Create a single source of truth for calculated KPIs and threshold-based alerts.
- Demonstrate a scalable architecture capable of handling at least 50,000 transactional records in the initial prototype.

4.3 Success Metrics

Metric	Target for prototype	Measurement
API response time	<= 500 ms for common aggregated dashboard queries	95th percentile under representative indexed workload
Dashboard usability	Key KPIs visible within one screen at desktop resolution	UX review / acceptance test
Data generation	50,000 shipments / 24 months	Record-count validation
RBAC enforcement	0 cross-scope data leakage in security tests	Automated authorization test suite
Anomaly detection	Known injected anomalies detected	Seeded anomaly test cases
TRS accuracy	Calculated duration matches source timestamps	Automated formula validation

5. Stakeholders and User Classes

Role	Primary responsibilities	Data scope	Typical actions
National Auditor	National oversight, comparative performance,	All ports / national	Compare ports, inspect alerts, export reports

Role	Primary responsibilities	Data scope	Typical actions
	review		
National Customs Analyst	KPI analysis and trend interpretation	All ports / national	Analyze revenue, TRS, enforcement, trends
Port Manager	Operational management for assigned terminal(s)	Assigned port(s) only	Review TRS, bottlenecks, anomalies, local alerts
Revenue Officer	Revenue and valuation analysis	Authorized port/national scope	Review theoretical revenue gap and valuation anomalies
Enforcement Officer	Risk and enforcement performance	Authorized scope	Review flagged/seized outcomes, anomaly lists
System Administrator	Configuration and administration	All system metadata	Manage users, roles, rules, ports, KPI thresholds
Executive Viewer	High-level management reporting	Read-only national / assigned scope	View executive KPI scorecard and trends

5.1 Role-to-Permission Matrix

Role	Dashboard	TRS	Revenue	Enforcement	Org Development	Alerts	Rules	Users	Exports	Audit Logs
National Auditor	R	R	R	R	R	R	RW	-	R	R
National Customs Analyst	R	R	R	R	R	R	R	-	R	R
Port Manager	R	RW	R	R	R	R	-	-	R	-
Revenue Officer	R	R	RW	R	-	R	-	-	R	-
Enforcement Officer	R	R	R	RW	-	R	-	-	R	-
System Administrator	R	R	R	R	R	RW	RW	RW	RW	R
Executive Viewer	R	R	R	R	R	R	-	-	R	-

Legend: R = Read, RW = Read/Write, - = No access. Final permissions must also respect port/terminal scope.

6. Scope

6.1 In Scope

- Authentication and JWT-based authorization.
- Role-based access control with port-level and national-level data segregation.
- Executive dashboard with KPI cards, trends, comparisons, and status indicators.
- Trade Facilitation / TRS analytics based on arrival and release timestamps.
- Revenue analytics using actual tax, expected tariff rate, historical reference value, and analytical revenue gap.
- Enforcement analytics for inspection status, flagged declarations, and seizures.
- Organizational Development scorecard using configurable operational/administrative KPIs.
- Configurable thresholds and UI alerts.
- Shipment-level valuation anomaly table.
- Date and port filtering with dynamic refresh.
- CSV/XLSX/PDF-ready export architecture (CSV mandatory in prototype; other formats optional).
- Audit logging for security-sensitive and configuration actions.
- Synthetic data generation with deliberate anomalies for repeatable testing.

6.2 Out of Scope for the Initial Prototype

- Live integration with a national customs declaration system or production customs single window.
- Official tariff determination or legal customs valuation decisions.
- Real-time vessel/aircraft GPS tracking.
- Direct payment processing or banking integration.
- Biometric identity management.
- Machine-learning models trained on real customs data.
- Automated enforcement decisions or automatic seizure orders.

7. Assumptions, Constraints, and Dependencies

7.1 Assumptions

- The shipment dataset is simulated and may intentionally contain anomalies for demonstration.
- The system will start with a fixed 24-month data window but must allow future dates and additional records.
- Each shipment has one primary port and one primary HS code in the simplified transactional model.
- Arrival time precedes or equals release time for valid released shipments.
- The historical average value is an analytical reference benchmark, not an official customs valuation determination.
- Tax rates are stored as configurable master data and may vary by jurisdiction in a production extension.

7.2 Technical Constraints

- Primary relational database: MySQL 8.x or compatible managed MySQL service.
- Backend: Node.js with Express.js and JWT middleware.
- Frontend: React + TypeScript + Vite. Next.js is optional only if deployment/routing requirements justify it; the baseline implementation shall remain React + TypeScript.
- Visualization: Recharts or Nivo; AG Grid for high-volume analytical tables.
- Styling: Tailwind CSS or MUI. Global state: Redux Toolkit or equivalent typed state layer for authentication and cross-module filters. Global state: Redux Toolkit or equivalent typed state layer for authentication and cross-module filters.
- Version control: Git/GitHub. Frontend and backend source shall use TypeScript with strict type checking where applicable. Frontend and backend source shall use TypeScript with strict type checking where applicable.
- API style: REST/JSON with typed request/response DTOs and centralized API contract types.
- Target desktop viewport for first release: 1366 x 768 and above; responsive support required for tablet layouts.

7.3 Dependencies

- Node.js runtime and package registry access.
- MySQL server and credentials.
- JWT secret or signing key.
- Seed/configuration datasets for ports, HS codes, users, roles, and KPI rules.
- WCO reference material for conceptual PMM mapping and TRS methodology.

8. Functional Requirements

8.1 Authentication and Session Management

FR-AUTH-001 | Login | Must
The system shall provide a username/email and password login form and authenticate against the application user store.

FR-AUTH-002 | JWT issuance | Must
On successful authentication, the backend shall issue a signed JWT containing user identity, role, and/or role reference plus an expiry time.

FR-AUTH-003 | JWT validation | Must
Protected API routes shall reject missing, malformed, expired, or invalid tokens with HTTP 401.

FR-AUTH-004 | Authorization | Must
The system shall apply role and scope checks before executing queries that return protected data.

FR-AUTH-005 | Logout | Must
The frontend shall clear session state on logout. Token invalidation/rotation may be added through refresh-token strategy in production.

FR-AUTH-006 | Password storage | Must
Passwords shall never be stored in plaintext and shall be hashed using a modern password hashing algorithm such as Argon2id or bcrypt.

8.2 Executive Dashboard

FR-DASH-001 | KPI cards | Must
Display configurable summary KPIs including YTD revenue, average clearance time, shipment volume, anomaly count, flagged count, and seizure count.

FR-DASH-002 | Trend charts | Must
Display time-series trends at monthly or daily granularity depending on selected date range.

FR-DASH-003 | Port comparison | Must
Allow authorized users to compare selected ports on key indicators.

FR-DASH-004 | KPI status | Must
Display Normal, Warning, and Critical state based on configured thresholds or target ranges.

FR-DASH-005 | Drill-down | Must
Clicking a KPI/chart element shall navigate to a filtered analytical detail view.

FR-DASH-006 | Global filters | Must
Selected date range, port, location type, HS code, and inspection status shall be consistently applied across supported dashboard modules.

FR-DASH-007 | Executive scorecard | Must
Display Total Revenue (YTD), Analytical Revenue Gap %, National Average Clearance Time, Seizure Rate, Open Alerts, and shipment volume as the default executive scorecard.

FR-DASH-008 | Live refresh indicator | Should
Display the last data refresh timestamp and support configurable polling (prototype default 60 seconds) without requiring a full page reload.

8.3 Trade Facilitation / TRS

FR-TRS-001 | Average release time | Must
Calculate average elapsed time between arrival_time and release_time for valid released shipments.

FR-TRS-002 | Port aggregation | Must
Group TRS metrics by port, month, and optionally location_type.

FR-TRS-003 | Distribution metrics | Must
Provide at least average, median, minimum, maximum, and percentile-capable architecture; P50/P90 recommended.

FR-TRS-004 | Bottleneck segmentation | Must
Allow analysis by port and time period; future extension may include process-stage timestamps.

FR-TRS-005 | Outlier handling | Must
Support configurable outlier rules; abnormal negative durations shall be excluded and logged as data-quality errors.

FR-TRS-006 | Target comparison | Must
Compare port or national averages against configurable targets such as 72 hours.

FR-TRS-007 | Release-time distribution buckets | Should
Display release-time distribution buckets such as <6h, 6-24h, 24-72h, and >72h to complement mean/median/P90 metrics.

FR-TRS-008 | Bottleneck ranking | Should
Rank ports by average release time and identify configured bottleneck ports for management review.

8.4 Revenue Collection

FR-REV-001 | Actual revenue | Must
Calculate total tax_collected over the selected filter scope.

FR-REV-002 | Reference value | Must
Retrieve hs_codes.historical_avg_value for analytical comparison with declared value.

FR-REV-003 | Theoretical revenue | Must
Calculate an estimated theoretical revenue using the configurable analytical reference value multiplied by the standard tax rate.

FR-REV-004 | Revenue gap | Must
Calculate the non-negative difference between theoretical revenue and actual tax collected for reporting purposes.

FR-REV-005 | Valuation anomaly | Must
Flag records whose declared value deviates from the historical reference value by a configurable percentage threshold.

FR-REV-006 | Anomaly detail | Must
Provide shipment ID, port, HS code, declared value, reference value, deviation percentage, tax collected, estimated revenue, and gap.

FR-REV-007 | Revenue-gap ranking | Should
Provide a ranked view of the top 10 HS codes and/or ports contributing the largest analytical revenue gaps.

FR-REV-008 | Secondary anomaly signal | Could
Optionally calculate a z-score of declared value within an HS-code peer group and expose it as a secondary review signal without replacing the primary configurable deviation rule.

8.5 Enforcement and Security

FR-ENF-001 | Status counts | Must
Aggregate shipments by inspection_status.

FR-ENF-002 | Flag rate | Must
Calculate flagged shipments as a proportion of total shipments in scope.

FR-ENF-003 | Seizure rate | Must
Calculate seized shipments as a proportion of total shipments in scope.

FR-ENF-004 | Trend analysis | Must
Display monthly flagged/seized trends.

FR-ENF-005 | Port comparison | Must
Compare enforcement outcomes by port with appropriate volume normalization.

FR-ENF-006 | Normalized interception effectiveness | Should
Provide normalized enforcement views so high-volume ports are not interpreted as inherently higher risk solely because they process more declarations.

8.6 Organizational Development

FR-ORG-001 | Operational scorecard | Must
Provide an extensible scorecard for organizational indicators such as data completeness, alert response time, report freshness, system availability, or training completion where those datasets are configured.

FR-ORG-002 | Targets | Must
Allow administrators to configure target values and threshold rules for organizational KPIs.

FR-ORG-003 | Trend view | Must
Show organizational KPI performance over time.

FR-ORG-004 | Scope enforcement | Must
Apply the same RBAC and port scoping rules where organizational indicators are location-specific.

FR-ORG-005 | Per-port monthly operational indicators | Should
Store and display officers on duty, training hours, system uptime percentage, and backlog count by port and month when those datasets are configured.

8.7 Dynamic Rule Engine and Alerts

FR-RULE-001 | Rule definition | Must
Administrators shall define rules with KPI name, operator, threshold, severity, scope, active dates, and enabled status.

FR-RULE-002 | Continuous evaluation | Must
Rules shall be evaluated whenever relevant analytics are refreshed or at a configurable scheduled interval.

FR-RULE-003 | Alert creation | Must
A breached rule shall create an alert record with a timestamp, affected scope, observed value, threshold, and status.

FR-RULE-004 | Severity | Must
Supported severities shall include Info, Warning, and Critical.

FR-RULE-005 | Acknowledgement | Must
Authorized users shall be able to acknowledge alerts without changing the underlying KPI data.

FR-RULE-006 | Deduplication | Must
The engine shall avoid generating duplicate alerts for the same rule, scope, KPI, and evaluation window unless a configurable re-alert interval has elapsed.

FR-RULE-007 | Scheduled evaluation | Must
Evaluate active rules on a configurable scheduler; prototype default is every 5 minutes over a rolling evaluation window.

FR-RULE-008 | Rule editor permissions | Must
Authorized Rule Managers shall create, edit, enable, and disable KPI rules. The prototype shall permit National Auditors and System Administrators to perform rule management, subject to audit logging.

FR-RULE-009 | Auto-resolution | Should
An alert may automatically transition to Resolved when the monitored KPI returns to the configured compliant range, while preserving the full alert history.

FR-RULE-010 | Alert panel | Must
The notification panel shall display unread/open counts and shall enforce the same port scope as the underlying KPI.

8.8 Administration

FR-ADM-001 | User management | Must
Administrators shall create, deactivate, and update users subject to authorization constraints.

FR-ADM-002 | Role assignment | Must
Administrators shall assign roles to users.

FR-ADM-003 | Port scope | Must
Administrators shall associate Port Managers and scoped users with one or more permitted ports.

FR-ADM-004 | KPI configuration | Must
Administrators shall maintain KPI targets, thresholds, and display status rules.

FR-ADM-005 | Master data | Must
Administrators shall manage ports and HS code reference records through protected CRUD operations.

FR-ADM-006 | Audit log | Must
Administrative changes shall be written to the audit log.

8.9 Search, Filter, Drill-Down, and Export

FR-UX-001 | Date range | Must
The user shall be able to select a start and end date.

FR-UX-002 | Port selector | Must
The port selector shall show only ports authorized for the current user.

FR-UX-003 | HS filter | Must
Authorized users shall be able to filter by HS code or HS description.

FR-UX-004 | Server-side filtering | Must
For large datasets, filter parameters shall be passed to the API and evaluated in SQL rather than downloaded and filtered in the browser.

FR-UX-005 | Export | Must
The system shall provide export of visible analytical tables to CSV; XLSX/PDF may be added as phase-two enhancements.

FR-UX-006 | Empty states | Must
Charts and tables shall show a clear no-data state rather than blank components.

9. PMM Dimension Requirements

Dimension mapping
The following module labels are intentionally aligned to the current WCO PMM dimension wording while preserving the simplified four-module structure proposed for the project. [1]

WCO PMM dimension	Dashboard module	Core prototype measures
Trade Facilitation and Economic Competitiveness	Trade Facilitation / TRS	Average release time, median release time, P90, volume, target compliance

WCO PMM dimension	Dashboard module	Core prototype measures
Revenue Collection	Revenue Intelligence	Actual tax, reference value variance, theoretical revenue, analytical revenue gap
Enforcement, Security and Protection of Society	Enforcement & Security	Flagged, cleared, seized, flag/seizure rates, trends
Organizational Development	Organizational Scorecard	Configurable organizational KPIs, target compliance, alert-response or data-quality measures

9.1 Trade Facilitation and Economic Competitiveness

The trade facilitation module shall emphasize cargo release speed, predictability, and comparison across entry points. The WCO describes TRS as a methodology for measuring the time required from arrival of goods to physical release and using the results to identify bottlenecks and improvement measures. The current WCO site lists TRS Guide Version 4 (2025). [2]

- Measure elapsed release time in hours.
- Aggregate monthly and port-level statistics.
- Highlight ports exceeding configured targets.
- Show trend direction and period-over-period change.
- Allow drill-down from a national metric to port and shipment evidence.

9.2 Revenue Collection

The revenue module uses a simulated analytical benchmark. It is specifically intended to demonstrate data analytics rather than to replace statutory customs valuation methods. The benchmark is derived from historical averages stored in the HS code master data and is combined with a configurable standard tax rate.

9.3 Enforcement, Security and Protection of Society

The enforcement module aggregates operational outcomes and provides normalized indicators so that a high-volume port is not automatically interpreted as a higher-risk port solely because it has more declarations.

9.4 Organizational Development

The organizational-development module is deliberately extensible. Because the simplified transactional schema does not yet contain staff, training, process maturity, or technology service data, the initial implementation shall use configurable KPI records rather than inventing unsupported operational facts.

10. Advanced Analytics Requirements

10.1 Predictive Valuation and Revenue Analytics

For each shipment, the system shall compare the declared value with the historical reference value associated with the HS code. The calculation is an analytical anomaly-detection model and should not be presented as a legally binding valuation assessment.

Metric	Formula	Purpose
Valuation deviation %	((Reference Value - Declared Value) / Reference Value) x 100	Measures how far declared value falls below/above the analytical benchmark.
Estimated theoretical revenue	Reference Value x (Standard Tax Rate / 100)	Estimates revenue using the benchmark instead of the declared value.
Revenue gap	MAX(0, Estimated Theoretical Revenue - Actual Tax Collected)	Highlights an analytical shortfall.
Anomaly flag	Deviation % >= configurable threshold	Identifies candidate records for review.

Example

If a simulated HS code has a reference value of 100,000, a declared value of 50,000, and a standard tax rate of 10%, the dashboard can show a 50% valuation deviation and an analytical theoretical-revenue estimate of 10,000. The record is then flagged when the configured anomaly threshold is <= 50%.

10.2 Time-Release Study Analytics

The TRS module shall compute release durations from timestamps and support monthly, port, and national views. The WCO emphasizes segmentation, structured time-interval analysis, trend analysis, and outlier management in its updated TRS methodology. [2]

Metric	Definition	Display
Average clearance time	Arithmetic mean of valid durations	KPI card + line chart
Median / P50	50th percentile of valid durations	KPI card/table
P90	90th percentile of valid durations	KPI card/table
Fastest release	Minimum valid duration	Detail/table
Slowest release	Maximum valid duration	Detail/table
Target compliance	% of releases within target hours	KPI/benchmark

10.3 Dynamic Rule Engine

Field	Description	Example
rule_id	Unique rule identifier	RULE-TRS-001
metric_code	Metric being monitored	AVG_CLEARANCE_HOURS
operator	Comparison operator	>
threshold_value	Numeric threshold	72
scope_type	National or port	PORT
scope_id	Applicable scope identifier	12
severity	Info/Warning/Critical	Critical
active_from / active_to	Rule validity period	2026-01-01 to NULL
enabled	Activation flag	true

11. Business Rules and KPI Definitions

Rule ID	Business rule
BR-001	A released shipment must have release_time >= arrival_time; invalid records are excluded from performance calculations and written to a data-quality log.
BR-002	A valuation anomaly threshold is configurable; the prototype default is 30% and test fixtures must include at least one 50% anomaly.
BR-003	Theoretical revenue is an analytical estimate and must be labeled as such in the UI.
BR-004	Port-scoped users may only query records belonging to their authorized ports.
BR-005	National users may view all authorized national entry points.
BR-006	KPIs must show the selected date range and filter scope to prevent interpretation errors.
BR-007	Threshold alerts are created from configured rules, not hard-coded UI conditions.
BR-008	Shipment-level financial values are rounded to two decimal places for display; aggregation uses full database precision.
BR-009	Empty result sets return valid JSON with zero/empty collections and a descriptive message where appropriate.
BR-010	Exported data must apply the same authorization scope as interactive queries.

11.1 KPI Dictionary

KPI Code	KPI	Definition	Unit	Direction
KPI-TRS-01	Average Release Time	Mean elapsed time from arrival to release	Hours	Lower is better
KPI-TRS-02	P90 Release Time	90th percentile elapsed time	Hours	Lower is better
KPI-TRS-03	Target Compliance	Releases completed within configured target	%	Higher is better
KPI-REV-01	Actual Revenue	Sum of tax collected	Currency	Higher is context-dependent
KPI-REV-02	Valuation Anomaly Count	Number of declarations exceeding deviation threshold	Count	Lower is typically better
KPI-REV-03	Analytical Revenue Gap	Estimated benchmark revenue less actual tax collected when positive	Currency	Lower is better
KPI-ENF-01	Flag Rate	Flagged shipments / total shipments	%	Context-dependent
KPI-ENF-02	Seizure Rate	Seized shipments / total shipments	%	Context-dependent
KPI-ORG-01	Data Completeness	Valid populated records / expected records	%	Higher is better

12. User Interface and UX Requirements

12.1 Global Layout

- Persistent left sidebar with four PMM dimension modules and administration options based on role.
- Top header containing application title/breadcrumbs, global date filter, port selector, user identity, role badge, and notifications.
- Main workspace using responsive cards, charts, and tables.
- Consistent status language: Normal, Warning, Critical.
- All charts must provide tooltips and a visible legend where multiple series are present.

12.2 Required Screens

Screen	Primary content	Key interactions
Login	Authentication form, error messages	Login, show/hide password, validation
National Dashboard	KPI cards, trend charts, port comparison, active alerts	Date/port filters, drill-down
TRS Analysis	Release-time trend, port comparison, distribution summary	Time period, port, export
Revenue Intelligence	Revenue cards, anomaly chart, anomaly table	HS filter, threshold, drill-down
Enforcement	Flag/seizure counts, rates, trends	Status/port/date filters
Organizational Scorecard	Configured operational KPIs and target status	KPI selection, period filter
Alerts Center	Open/acknowledged/closed alerts	Filter, acknowledge, view rule
Administration	Users, ports, HS codes, KPI rules, thresholds	CRUD subject to RBAC
Audit Log	Administrative and security events	Date/action/user filters

12.3 Loading, Empty, and Error States

12.4 AG Grid Data-Grid Requirements

- Use AG Grid as the standard tabular component for shipment data, valuation anomalies, alerts, audit logs, users, and other high-volume administrative/detail lists.
- AG Grid shall support server-backed pagination, sorting, filtering, column resizing, column visibility controls, row selection, and CSV export for permitted datasets.
- Filters applied in AG Grid shall map to backend query parameters; authorization and filtering shall be enforced in SQL/API services rather than relying on browser-only filtering.
- Grid definitions shall be strongly typed with TypeScript row models, column definitions, and response DTOs. Avoid untyped any values except documented integration boundaries.
- Financial columns shall use appropriate numeric formatting; timestamps shall render consistently with the system timezone policy while source timestamps remain UTC.
- Anomaly grids shall visually distinguish flagged severity/review state using text, icons, or badges so status does not depend on color alone.
- Grid loading shall use skeleton/overlay states; empty datasets shall show a descriptive empty state; API failures shall show retry actions.
- Use skeleton cards and chart placeholders for initial loading.
- Use inline retry controls for recoverable API failures.
- Use explicit empty-state messaging when filters produce no records.
- Never show raw SQL errors or stack traces to end users.
- Show the last refresh timestamp for data-driven dashboard views.

13. System Architecture

13.1 Three-Tier Architecture

Layer	Technology	Responsibilities
Presentation	React + TypeScript + Vite + Tailwind/MUI + AG Grid + Recharts/Nivo	Typed UI, routing, global filter state, role-aware navigation, analytical tables, charts, accessibility
Application	Node.js + Express.js + TypeScript	Authentication, authorization, validation, business rules, aggregation API orchestration, scheduled jobs, audit events
Data	MySQL 8.x	Relational storage, indexing, joins, GROUP BY aggregations, transactional integrity

13.2 Logical Architecture

Component	Inputs	Outputs	Notes
Auth Controller	Credentials	JWT / auth result	JWT-based session security
RBAC Middleware	JWT claims + route + scope	Authorized request context	Must execute before protected queries
KPI Service	Filters + user scope	Aggregated KPI JSON	Centralizes KPI formulas
TRS Service	Shipment timestamps	TRS metrics	Uses indexed date/port columns
Revenue Service	Shipments + HS master	Revenue and anomaly metrics	Analytical, not legal valuation
Enforcement Service	Inspection statuses	Counts/rates/trends	Volume-normalized views

Component	Inputs	Outputs	Notes
Rule Engine	KPI results + rules	Alerts	Supports severity/deduplication
Audit Service	Action context	Audit records	Captures admin/security changes
MySQL	Transactional + master data	Query results	Primary system of record for prototype

13.3 Data Flow

```
User -> React UI -> Axios/fetch -> Express Route -> JWT/RBAC Middleware -> KPI/Domain Service -> SQL
Query -> MySQL
                                                |
                                                +-> Audit / Rule Evaluation -> Alert Store

MySQL Aggregation -> JSON KPI Payload -> React State -> Charts / Tables / KPI Cards
```

13.4 Recommended Project Structure

13.5 TypeScript Implementation Standards

- Frontend source shall use .ts and .tsx files with strict compiler settings enabled where compatible with dependencies.
- Backend controllers, services, repositories, validators, jobs, and route handlers shall use TypeScript types/interfaces for request payloads, query parameters, domain objects, and service outputs.
- API response DTOs used by the React client shall be represented by shared TypeScript types or generated/validated contracts so KPI cards, charts, and AG Grid rows remain synchronized with backend payloads.
- Use discriminated unions/enums for role names, alert status, severity, inspection status, and KPI state values where practical.
- ESLint and Prettier shall enforce consistent TypeScript/React style; CI shall fail on type errors and lint failures.
- Use a typed API service layer instead of calling fetch/Axios directly from every page component.

```
pmm-dashboard/
 backend/
  src/
    config/
    controllers/
    middleware/
    routes/
    services/
    repositories/
    validators/
    jobs/
    utils/
    app.ts
  sql/
    schema.sql
    indexes.sql
    seed.sql
  tests/
 frontend/
  src/
    components/
    pages/
    layouts/
    hooks/
    context/
    services/
    utils/
    charts/
    types/
    grids/
    state/
  public/
    main.tsx
 data-generator/
```

```
  generate_shipments.py
docs/
  SRS
  API
README.md
```

14. Data Architecture and Database Design

14.1 Core Entities

Entity	Purpose	Cardinality
ports	Customs entry points / terminals	1 port -> many shipments
hs_codes	Commodity classification and analytical tariff/reference data	1 HS -> many shipments
shipments	Primary transactional customs declaration/cargo record	Many per port and HS code
users	Application identities	Many users
roles	Authorization definitions	Many users -> one or more roles, implementation may start one role/user
user_ports	Port-level access scope	Many-to-many users/ports
kpi_rules	Configurable KPI thresholds	Many rules per KPI/scope
alerts	Generated threshold breach records	Many alerts per rule
audit_logs	Security/admin traceability	Many events

14.2 Core Table: ports

Column	Type	Constraints	Description
port_id	INT	PK, AUTO_INCREMENT	Unique port identifier
port_name	VARCHAR(100)	NOT NULL, UNIQUE	Customs entry point name
location_type	VARCHAR(50)	NOT NULL	Seaport, Airport, Land Border
is_active	BOOLEAN	DEFAULT TRUE	Soft availability flag
created_at	DATETIME	NOT NULL	Record creation time
updated_at	DATETIME	NOT NULL	Last modification time

14.3 Core Table: hs_codes

Column	Type	Constraints	Description
hs_code	VARCHAR(10)	PK	HS master identifier; prototype may use 6-digit codes
description	TEXT	NOT NULL	Commodity description
standard_tax_rate	DECIMAL(5,2)	NOT NULL	Analytical standard tariff/tax percentage
historical_avg_value	DECIMAL(15,2)	NOT NULL	Analytical historical reference value
currency_code	CHAR(3)	DEFAULT configurable currency	Currency of reference value
is_active	BOOLEAN	DEFAULT TRUE	Master data status

HS code note

The WCO Harmonized System uses a six-digit international commodity classification structure, with national tariff schedules potentially extending or adapting classification and rates. For this prototype, the HS master is synthetic and should not be represented as an authoritative national tariff schedule. [4]

14.4 Core Table: shipments

Column	Type	Constraints	Description
shipment_id	VARCHAR(50)	PK	Unique declaration/tracking identifier
port_id	INT	FK -> ports.port_id, INDEX	Customs point
hs_code	VARCHAR(10)	FK -> hs_codes.hs_code, INDEX	Commodity classification
declared_value	DECIMAL(15,2)	NOT NULL	Importer-declared value
tax_collected	DECIMAL(15,2)	NOT NULL	Actual tax collected in simulated data
arrival_time	DATETIME	INDEX, NOT NULL	Cargo arrival timestamp
release_time	DATETIME	INDEX, NULLABLE	Cargo release timestamp; null when not released
inspection_status	VARCHAR(20)	INDEX, NOT NULL	Cleared, Flagged, Seized
created_at	DATETIME	NOT NULL	Ingestion time

14.5 Supporting Tables

Table	Key fields	Purpose
users	user_id, email, password_hash, role_id, status	Identity and authentication
roles	role_id, role_name	Role catalog
user_ports	user_id, port_id	Port-scope assignment
kpi_rules	rule_id, metric_code, operator, threshold_value, scope_type, scope_id, severity, enabled	Dynamic thresholds
alerts	alert_id, rule_id, metric_code, observed_value, threshold_value, scope_id, severity, status, created_at, acknowledged_at	Alert lifecycle
audit_logs	audit_id, user_id, action, entity_type, entity_id, metadata_json, ip_address, created_at	Traceability
kpi_targets	target_id, metric_code, target_value, unit, period_type, scope_type, scope_id	KPI target library
org_metrics	port_id, metric_month, officers_on_duty, training_hours, system_uptime_pct, backlog_count	Monthly per-port organizational and operational indicators

14.6 Indexing Strategy

Index	Columns	Reason
IDX_SHIP_ARRIVAL	shipments(arrival_time)	Date-range filtering and TRS aggregation
IDX_SHIP_PORT_ARR	shipments(port_id, arrival_time)	Port + date dashboard filters
IDX_SHIP_HS_ARR	shipments(hs_code, arrival_time)	HS + time revenue analytics
IDX_SHIP_STATUS_ARR	shipments(inspection_status, arrival_time)	Enforcement counts by period
IDX_PORT_NAME	ports(port_name)	Port lookup
IDX_USER_EMAIL	users(email)	Login lookup
IDX_ALERT_STATUS	alerts(status, created_at)	Active alert center
IDX_SHIP_PORT_STATUS_ARR	shipments(port_id, inspection_status, arrival_time)	Scoped enforcement tables and AG Grid detail feeds

Performance caveat

The 500 ms value is a design target, not a guarantee. Validate it with representative load testing before release.

15. API Requirements

15.1 API Conventions

All analytical list endpoints consumed by AG Grid shall support server-side pagination, sorting, and filtering through validated query parameters such as page, pageSize, sortBy, sortDir, filterModel, startDate, endDate, portId, hsCode, and inspectionStatus. The backend shall return totalRows or an equivalent pagination total so the grid can render accurate navigation.

For the prototype, AG Grid shall use a server-backed/infinite data access pattern so large tables do not require full dataset downloads. Enterprise-only AG Grid features shall not be required for baseline acceptance.

- Base path: /api/v1
- Transport: HTTPS in non-local environments.
- Content type: application/json unless a file export endpoint is used.
- Authentication: Authorization: Bearer <JWT>.
- Dates: ISO 8601 strings; server shall normalize timezone handling consistently.
- Pagination: limit + offset or cursor for large detail lists.
- Error payload: { code, message, details?, requestId }.

15.2 Authentication Endpoints

Method	Endpoint	Purpose
POST	/api/v1/auth/login	Authenticate and issue access token
POST	/api/v1/auth/refresh	Optional refresh-token flow
POST	/api/v1/auth/logout	Client/session termination
GET	/api/v1/auth/me	Return current authenticated user and role/scope

15.3 KPI Endpoints

Method	Endpoint	Key query parameters	Response
GET	/api/v1/metrics/summary	startDate, endDate, portId	Executive KPI summary
GET	/api/v1/metrics/trs	startDate, endDate, portId, hsCode	Monthly TRS metrics
GET	/api/v1/metrics/revenue	startDate, endDate, portId, hsCode	Revenue and valuation analytics
GET	/api/v1/metrics/enforcement	startDate, endDate, portId	Flag/seizure counts and rates
GET	/api/v1/metrics/org	startDate, endDate, portId	Organizational KPI scorecard

15.4 Detail Endpoints

Method	Endpoint	Purpose
GET	/api/v1/shipments/anomalies	List valuation anomaly candidates
GET	/api/v1/shipments/:id	Shipment detail
GET	/api/v1/ports	Authorized port list
GET	/api/v1/hs-codes	HS master search

15.5 Rules and Alerts Endpoints

Method	Endpoint	Purpose
GET	/api/v1/rules	List rules
POST	/api/v1/rules	Create rule
PATCH	/api/v1/rules/:id	Update rule
DELETE	/api/v1/rules/:id	Disable/delete rule
GET	/api/v1/alerts	List alerts

Method	Endpoint	Purpose
PATCH	/api/v1/alerts/:id/acknowledge	Acknowledge alert

15.6 Example TRS Response

```
GET /api/v1/metrics/trs?startDate=2025-01-01&endDate=2026-12-31&portId=3

{
  "filters": {"startDate":"2025-01-01","endDate":"2026-12-31","portId":3},
  "summary": {"averageHours":61.8,"medianHours":49.2,"p90Hours":118.7},
  "series": [
    {"month":"2026-01","averageHours":58.1,"shipmentCount":1820},
    {"month":"2026-02","averageHours":63.4,"shipmentCount":1751}
  ]
}
```

15.7 HTTP Status Codes

Code	Meaning	Typical use
200	Success	GET / metrics
201	Created	POST rule/user
204	No Content	Successful update without body
400	Bad Request	Validation failure
401	Unauthorized	Missing/invalid token
403	Forbidden	Role/scope violation
404	Not Found	Missing resource
409	Conflict	Duplicate master data
422	Unprocessable Entity	Business validation error
429	Too Many Requests	Rate limiting
500	Internal Server Error	Unexpected backend error

16. Security and Access Control

16.1 Security Requirements

SEC-001 | Least privilege | Must
Users shall receive only the permissions and data scope required by their role.

SEC-002 | Password security | Must
Passwords shall be hashed using a current password hashing algorithm; passwords shall never be logged.

SEC-003 | Token validation | Must
JWT signature and expiry shall be verified for every protected request.

SEC-004 | Input validation | Must
API inputs shall be schema-validated and sanitized before use.

SEC-005 | SQL injection prevention | Must
Use parameterized queries or a trusted query builder/ORM; never concatenate untrusted values into SQL.

SEC-006 | CORS | Must
Configure an allowlist for browser origins in non-local deployments.

SEC-007 | Secrets | Must
Database credentials and JWT signing keys shall come from environment variables or a secret manager.

SEC-008 | HTTPS | Must
All deployed traffic shall use HTTPS.

SEC-009 | Authorization before data access | Must
The backend shall derive permitted port scope from authenticated identity and not trust client-submitted role/scope fields.

SEC-010 | Sensitive logging | Must
Logs shall avoid passwords, full JWTs, and unnecessary sensitive financial payloads.

16.2 RBAC Enforcement Model

```
1. Verify JWT -> 2. Load user role/scope -> 3. Evaluate route permission -> 4. Constrain SQL by
authorized port IDs
-> 5. Execute aggregation -> 6. Return only authorized rows/metrics
```

17. Auditability, Logging, and Monitoring

Event	Log required?	Example
Successful login	Yes	user_id, timestamp, IP
Failed login	Yes	username/email hash or redacted identifier, timestamp, IP
Role change	Yes	actor, subject, old role, new role
Port-scope change	Yes	actor, subject, added/removed ports
Rule creation/update	Yes	rule ID, actor, change summary
Alert acknowledgement	Yes	user, alert ID, timestamp
Master-data change	Yes	entity and changed fields
Export action	Yes	user, report type, filters
API error	Yes	requestId, endpoint, status, sanitized error

17.1 Operational Observability

- Structured backend logs in JSON format for deployed environments.
- Unique requestId per API call.
- Health endpoint: GET /health.
- Readiness endpoint: GET /ready that verifies required dependencies.
- Database connection pool monitoring.
- Optional metrics endpoint compatible with Prometheus or a cloud monitoring service.

18. Non-Functional Requirements

Category	Requirement	Target / Standard
Performance	Common dashboard aggregations	P95 <= 500 ms under representative prototype load
Performance	Initial page render	Useful shell visible within 2 seconds on typical broadband environment; API-dependent data may load progressively
Scalability	Transactional volume	At least 50,000 shipments initially; schema/API should support 10x growth without redesign
Availability	Prototype uptime	Best effort; production target can be defined later
Security	RBAC/data isolation	0 unauthorized data accesses in acceptance security tests
Maintainability	Code organization	Separate controller/service/repository responsibilities
Reliability	Error handling	No unhandled promise rejection or database error sent to client
Usability	Desktop dashboard	Readable at 1366x768 and above
Accessibility	UI	Keyboard-accessible interactive controls;

Category	Requirement	Target / Standard
		meaningful labels and color-independent status indicators
Data integrity	Referential integrity	Enforced foreign keys for core entities
Observability	Diagnostics	Request IDs, structured logs, health checks

18.1 Availability and Recovery

- Database backups should be enabled for any shared/staging environment.
- Deployment must support rollback to the previous known-good build.
- Schema migrations must be versioned and tested before deployment.
- Seed/reference data must be repeatable from version-controlled scripts.

19. Error Handling and Resilience

19.1 Error Categories

Category	Example	System behavior
Validation	Invalid date range	HTTP 400/422 with field-level message
Authorization	Port Manager requests another port	HTTP 403; log security event
Authentication	Expired JWT	HTTP 401; frontend redirects to login/refresh
Data quality	release_time earlier than arrival_time	Exclude from KPI and log anomaly
Database	Transient connection failure	Return HTTP 503; retry only for safe idempotent operations
Unexpected	Unhandled server exception	HTTP 500 with requestId; detailed error only in server logs

19.2 Data Quality Rules

- Shipment IDs must be unique.
- Port and HS code foreign keys must exist.
- Declared value and tax collected must be non-negative in synthetic baseline data.
- Released shipments must have release_time populated.
- Unreleased shipments may have release_time = NULL and must be excluded from release-time calculations.
- Inspection status must belong to the allowed status vocabulary unless a master-status table is introduced.

20. Reporting and Export Requirements

Report	Contents	Format	Access
National KPI Snapshot	Selected-period summary metrics	CSV; PDF optional	National roles / authorized executive users
TRS Performance Report	Port/month release metrics and target variance	CSV	Authorized analytics users
Revenue Anomaly Report	Shipment-level valuation anomalies	CSV	Revenue/audit/admin per scope
Enforcement Report	Flagged/seized/cleared counts and rates	CSV	Enforcement/national roles per scope
Alerts Report	Alert history and acknowledgement state	CSV	Authorized users
Audit Log Report	Security/admin events	CSV	System Administrator / National Auditor as configured

20.1 Export Controls

- Exports must use the same authorization filters as interactive screens.
- Export metadata should include report name, generated timestamp, user, selected period, and scope.
- Large exports should be streamed or queued rather than building excessively large in-memory arrays.

21. Data Generation and Simulation Requirements

21.1 Dataset Target

Dataset	Target volume	Notes
Ports	5-20	Mix of seaports, airports, land borders
HS codes	100-500	Synthetic reference values and rates
Shipments	50,000	Distributed over 24 months
Users	20-100	Multiple roles and port assignments
Rules	10-50	Mix of national and port-specific thresholds

21.2 Data Distribution Rules

- Use repeatable random seeds so test environments can reproduce the same dataset.
- Spread shipment volume across months and ports with modest seasonality.
- Generate realistic but non-identifying declaration values.
- Generate realistic release durations with most records inside target and a smaller tail of slow releases.
- Generate inspection statuses with controlled proportions for Cleared, Flagged, and Seized.

21.3 Deliberate Anomaly Injection

21.4 Enhanced Simulation Controls

- All timestamps shall be stored and generated in UTC. “Real-time” prototype behavior means filter-triggered refresh plus optional polling; the prototype default dashboard polling interval is 60 seconds.
- The generator shall create at least 50,000 shipments across at least 6 simulated ports and 24 months of activity.
- The generator shall inject approximately 3-5% valuation anomalies with declared value materially below the HS historical benchmark; at least one deterministic fixture shall be 50% below the benchmark for acceptance testing.
- Port-specific release-time distributions shall create at least one intentionally slow/bottleneck port with average release time above the configured target in selected months.
- The generator shall seed organizational indicators per port/month where enabled: officers on duty, training hours, system uptime percentage, and backlog count.
- A fixed random seed shall be supported so the same test dataset can be regenerated deterministically.
- At least 20 seeded valuation anomalies where declared_value is 50% or more below historical_avg_value.
- At least 20 long-release shipments above 72 hours.
- At least 10 records with inspection_status = Seized.
- At least 5 invalid timestamp records reserved for data-quality tests and excluded from KPI calculations.

21.4 Example Generation Pseudocode

```
for shipment in range(50000):
    port = weighted_choice(ports)
    hs = weighted_choice(hs_codes)
    ref = hs.historical_avg_value
    declared = random_value_around(ref)
    if shipment_id in seeded_anomaly_ids:
```

```
        declared = ref * 0.50
    arrival = random_datetime(month_window)
    release = arrival + random_duration()
    status = weighted_choice([Cleared, Flagged, Seized])
    tax = declared * hs.standard_tax_rate / 100
    insert(shipment)
```

22. Testing and Quality Assurance

22.1 Test Levels

Level	Coverage	Tools / Approach
Unit tests	Formula functions, validators, utilities	Jest/Vitest/Python pytest
API tests	Routes, auth, RBAC, error responses	Jest/Supertest/Postman/Newman
Integration tests	API + MySQL queries	Test MySQL database / Docker
UI tests	Filters, navigation, rendering	React Testing Library / Playwright
Security tests	Token and scope isolation	Automated API authorization matrix
Performance tests	Aggregation latency and concurrency	k6/Artillery/JMeter
UAT	Business workflows and KPI interpretation	Checklist with expected results

22.2 Critical Acceptance Test Cases

TC ID	Scenario	Expected result
TC-001	Valid user login	JWT issued; dashboard accessible according to role
TC-002	Invalid password	401; no token issued
TC-003	Expired token	401; protected request rejected
TC-004	Port Manager requests unauthorized port	403; no records returned
TC-005	TRS calculation on known timestamps	Exact expected hours returned
TC-006	Shipment with 50% under-declared value	Anomaly flagged when threshold <= 50%
TC-007	72-hour threshold breached	Alert created at correct severity
TC-008	Duplicate rule evaluation	No duplicate alert within configured deduplication window
TC-009	Invalid release timestamp	Record excluded from TRS and logged for data quality
TC-010	CSV export by port-scoped user	Export contains only authorized port data

22.3 Definition of Done

- All Must requirements in this SRS are implemented or explicitly waived.
- Critical test cases pass.
- No known critical/high security defects remain.
- Dashboard calculations reconcile to direct SQL verification queries.
- Seeded anomalies are discoverable from the UI.
- README and environment setup are documented.
- Application can be started from a clean environment using documented steps.

23. Deployment and DevOps

23.1 Environments

Environment	Purpose
Local	Developer workstation; local MySQL / Docker MySQL
Development	Shared integration and feature testing

Environment	Purpose
Staging	Release candidate / UAT
Production / Demo	Portfolio or controlled demonstration deployment

23.2 Environment Variables

```
NODE_ENV=development
PORT=5000
DATABASE_URL=mysql://user:password@host:3306/pmm_dashboard
JWT_SECRET=replace_with_secret
JWT_EXPIRES_IN=1h
CORS_ORIGIN=http://localhost:5173
```

23.3 Containerization

Docker is optional for the first local prototype but recommended to make the Node.js API and MySQL service reproducible. A docker-compose configuration may provide backend, database, and optional frontend services. Database volumes must be persisted outside the container lifecycle for non-ephemeral environments.

23.4 CI/CD Pipeline

- On pull request: install dependencies, lint, unit tests, API tests, and build.
- On main branch merge: build deployable artifacts and run integration tests.
- Staging deployment: run migrations and smoke tests.
- Production/demo deployment: require successful staging checks and an explicit release step.

24. 14-Day Implementation Plan

Day	Phase	Deliverables	Exit criteria
1	Architecture & Data	GitHub repo, React+TypeScript+Vite frontend skeleton, TypeScript backend skeleton, MySQL schema, AG Grid baseline, base config	Repo builds; DB schema applied
2	Data Engineering	50k shipment generator, reference data, anomaly seed logic	Seed dataset generated reproducibly
3	Backend Foundation	Express + TypeScript app, DB pool, typed middleware, controllers/services/repositories, scheduled-job scaffold	Server starts; DB connection verified
4	Auth & RBAC	Login, JWT, roles, user-port scope, typed auth context, server-side authorization tests	Protected endpoints reject unauthorized scope
5	TRS API	/metrics/trs queries and formulas	Known timestamp tests pass
6	Revenue API	/metrics/revenue, anomaly query	Seeded anomalies appear correctly
7	Enforcement API	Status aggregations and rates	Counts reconcile with raw

Day	Phase	Deliverables	Exit criteria
			data
8	QA Backend	Error handling, Postman/Newman collection, query optimization	API test suite green
9	Frontend Foundation	React + TypeScript + Vite layout, Tailwind/MUI, Redux Toolkit/global filters, AG Grid integration, sidebar/header/routing	All pages route and render shell
10	Dashboard KPIs	Summary cards, typed API service layer, filter context, executive scorecard	KPIs match API responses
11	Charts	TRS line/distribution charts, revenue anomaly visuals, enforcement visuals, bottleneck ranking	Charts render with loading/empty states
12	Global Filters	Date/port/HS/status filters, AG Grid server-backed sorting/filtering/pagination, drill-down behavior	Filter changes refresh supported modules
13	Polish & QA	Skeleton loading, AG Grid anomaly table, alerts UI, auto-resolution display, accessibility/type/lint pass	Critical UI defects resolved
14	Documentation & Demo	README, architecture notes, deployment notes, demo video	Release candidate documented and demonstrable

24.1 Recommended Git Branching

```
main
  ├── develop
  ├── feature/auth-rbac
  ├── feature/trs-metrics
  ├── feature/revenue-analytics
  ├── feature/enforcement
  ├── feature/dashboard-ui
  └── feature/rules-alerts
```

25. Acceptance Criteria

Additional v2.0 acceptance criteria

- AG Grid anomaly and shipment tables must paginate/sort/filter using backend query parameters without downloading the full 50,000-row dataset into the browser.
- Frontend and backend build pipelines must complete with zero TypeScript compiler errors.

- TRS distribution buckets (<6h, 6-24h, 24-72h, >72h) must reconcile to the filtered valid release population.
- The seeded bottleneck port must rank above configured target in at least one test period and generate a corresponding threshold alert within the scheduler window.
- Rule auto-resolution must preserve the original alert record and transition status only after the KPI returns to compliance.
- Organizational monthly indicators must be visible by authorized port and must respect RBAC scope.
- At least one deterministic 50% valuation anomaly and the injected anomaly population must appear in the AG Grid anomaly view at the configured threshold.
- National Auditors must be able to manage rules in the prototype only through authorized UI/API paths; every create/update/enable/disable action must produce an audit log.

1. The application can be started from a documented clean setup and the database schema can be initialized from version-controlled SQL.
2. The system contains at least 50,000 reproducible simulated shipments spanning 24 months.
3. The frontend authenticates users through the backend and enforces role and port scope.
4. The four PMM-aligned dashboard dimensions are accessible through role-aware navigation.
5. TRS average release time and trend metrics reconcile to independently calculated SQL results.
6. Revenue analytics correctly compute analytical theoretical revenue, gap, and configurable valuation anomalies.
7. Enforcement counts and normalized rates reconcile to source records.
8. Threshold rules create visible alerts with correct severity and without unauthorized cross-scope exposure.
9. Port-scoped users cannot view or export data from ports they are not assigned to.
10. National users can compare authorized entry points.
11. The main dashboard remains responsive and usable at 1366x768.
12. Critical security, data-integrity, and calculation test cases pass.
13. Documentation includes architecture, setup, environment variables, API overview, and known limitations.

26. Risks and Mitigations

Risk	Impact	Mitigation
Misinterpretation of analytical valuation as legal valuation	High	Label as analytical reference/estimate; document limitations; do not present as an official duty assessment.
Slow aggregation queries	High	Composite indexes, EXPLAIN plans, pre-aggregation/caching if needed, pagination.
RBAC data leakage	Critical	Server-side scope enforcement, authorization integration tests, never trust client scope identifiers.
Poor synthetic data realism	Medium	Seed distributions, domain review, repeatable scenarios, anomaly fixtures.
Timezone inconsistency	Medium	Store timestamps consistently, document timezone policy, convert at API/UI boundaries.
Threshold noise / alert fatigue	Medium	Deduplication, severity rules, acknowledgement workflow, configurable re-alert windows.
Changing WCO reference material	Medium	Version PMM/TRS alignment in the product documentation and keep mappings configurable.
Scope expansion	High	Use this SRS as baseline; new integrations become separate change requests.

Additional v2.0 risks

- AG Grid feature/licensing mismatch — keep the baseline dependent only on Community-available functionality and server-backed pagination; gate Enterprise-only features explicitly.
- Frontend/back-end type drift — mitigate with shared DTO types or generated schema contracts and CI type-checking.
- Over-fetching detail rows — mitigate with API-side pagination, selective columns, composite indexes, and EXPLAIN-plan validation.

27. Future Enhancements

- Move from synthetic records to controlled ingestion pipelines for real customs data.
- Add event-level customs process timestamps for deeper TRS bottleneck analysis.
- Add machine-learning risk scoring for valuation, misclassification, or enforcement prioritization.
- Add national tariff schedules and versioned tariff effective dates.
- Add geospatial visualization by port and trade corridor.
- Add scheduled email or in-app alert delivery.
- Add multilingual UI and accessibility localization.
- Add data warehouse/star-schema reporting for very high volumes.
- Add SSO/OIDC integration for government identity providers.
- Add immutable or externally stored audit trails for regulated deployments.

28. References

The following sources were consulted to validate the WCO framing used in this SRS. The project remains a simulated dashboard and does not claim official WCO implementation status.

[1] World Customs Organization - Performance Measurement Mechanism / PMM About. https://pmm.wcoomd.org/about — Four dimensions, PMM history, expected outcomes and KPIs.

[2] World Customs Organization - Time Release Study Guide, Version 4. https://www.wcoomd.org/en/topics/facilitation/instrument-and-tools/tools/time-release-study.aspx — TRS methodology and 2025 Version 4 reference.

[3] World Customs Organization - WCO Performance Measurement Mechanism. https://www.wcoomd.org/en/topics/capacity-building/instrument-and-tools/pmm.aspx — PMM overview, dimensions and benefits.

[4] World Customs Organization - Harmonized System FAQ. https://www.wcoomd.org/en/topics/nomenclature/overview/harmonized_system_faq.aspx — HS structure and classification background.

[5] World Customs Organization - PMM Library. https://pmm.wcoomd.org/library — PMM dimension/outcome/KPI reference library.

[6] Project-supplied PMM Dashboard SRS PDF (14 pages, uploaded with this revision) — additional prototype features reviewed for inclusion: release-time distribution buckets, bottleneck ranking, rule scheduling, alert auto-resolution, organizational monthly indicators, deterministic data seeding, top-10 revenue-gap views, and optional z-score anomaly signal.

Additional source supplied in the project brief: “An Overview of Performance Measurement in Customs Administrations” (ResearchGate). The source can be retained as background reading, but official WCO pages are preferred for normative/current PMM framing.

29. Appendices

Appendix A - Sample SQL for TRS Aggregation

```
SELECT
  s.port_id,
  DATE_FORMAT(s.arrival_time, '%Y-%m') AS month,
  AVG(TIMESTAMPDIFF(MINUTE, s.arrival_time, s.release_time)) / 60 AS avg_release_hours,
  COUNT(*) AS shipment_count
FROM shipments s
WHERE s.release_time IS NOT NULL
  AND s.arrival_time >= ?
  AND s.arrival_time < ?
  AND s.arrival_time <= s.release_time
  AND s.port_id IN (...)
GROUP BY s.port_id, DATE_FORMAT(s.arrival_time, '%Y-%m')
ORDER BY month, s.port_id;
```

Appendix B - Sample SQL for Valuation Anomalies

```
SELECT
  s.shipment_id, s.port_id, s.hs_code,
  s.declared_value, h.historical_avg_value,
  ((h.historical_avg_value - s.declared_value) / NULLIF(h.historical_avg_value,0)) * 100 AS
deviation_pct,
  (h.historical_avg_value * h.standard_tax_rate / 100) AS theoretical_revenue,
  GREATEST(0, (h.historical_avg_value * h.standard_tax_rate / 100) - s.tax_collected) AS revenue_gap
FROM shipments s
JOIN hs_codes h ON h.hs_code = s.hs_code
WHERE ((h.historical_avg_value - s.declared_value) / NULLIF(h.historical_avg_value,0)) * 100 >= ?
  AND s.arrival_time >= ?
  AND s.arrival_time < ?;
```

Appendix C - Example API Error

```
{
  "code": "PORT_SCOPE_VIOLATION",
  "message": "The authenticated user is not authorized to access the requested port.",
  "requestId": "REQ-20261005-AB12CD"
}
```

Appendix D - Minimum Definition of a Production-Hardened Version

- Approved PMM version and KPI mapping documented against official WCO material.
- Production-grade SSO/MFA, secret management, centralized logging, backup, monitoring, and disaster recovery.
- Formal data governance and data retention policy.
- Validated national tariff/valuation business rules reviewed by customs domain experts.
- Load testing at expected concurrent-user and transaction volumes.
- Formal penetration testing and security sign-off.
- Operational runbook, incident response process, and change-management process.

End of SRS

This document is the baseline requirements specification for the Customs PMM Dashboard prototype. Any requirement that changes product scope, data semantics, architecture, or security behavior should be logged as a controlled change request and reflected in the next SRS revision.